PETER O'BRIEN

Department of Agricultural and Resource Economics
University of California, Davis

paobrien@ucdavis.edu
peter-obrien.com

Education

University of California, Davis

Ph.D. in Agricultural and Resource Economics — Sep 2022 – Jun 2027 (expected)
M.S. in Agricultural and Resource Economics — Awarded 2024

University of Maine

B.S. in Economics, Minor in Mathematics, *summa cum laude* — Sep 2018 – Dec 2021

Fields of Interest

Environmental Economics, International Trade, and Agricultural Economics

References

Katrina Jessoe **(Co-chair)**	James Sayre **(Co-chair)**	Ishan Nath
Associate Professor	Assistant Professor	Assistant Professor
Agricultural and Resource Economics	Agricultural and Resource Economics	Harvard Kennedy School
UC Davis	UC Davis	Harvard University
kkjessoe@ucdavis.edu	jsayre@ucdavis.edu	ishannath@hks.harvard.edu

Research in Progress

Feeding the Export Boom: Technological Change and Land Use in Brazil's Beef Industry
(**Job Market Paper**)

Abstract: In Brazil, expanding agricultural production has driven deforestation across carbon-rich ecosystems, including the Amazon. Rapid growth in beef exports raises concerns about further land conversion, yet access to foreign markets may also encourage agricultural intensification, or the adoption of technologies that reduce land requirements per unit of output. Because intensification is difficult to observe, I use satellite imagery and deep learning to build a panel of Brazilian feedlots, facilities where cattle are produced in confinement. I combine this panel with records of processing-plant authorizations to study the effects of Brazil's recent beef export boom to China. I exploit variation in the timing of export authorizations and regions' exposure to plants and find that greater export exposure substantially increased feedlot entry and total capacity. Holding current beef output fixed, replacing this export-induced intensive production with traditional, grazing-based systems would require additional pasture expansion and put 169 $MtCO_2$ of carbon stocks at risk, underscoring how trade-induced technological change can shape the environmental consequences of agricultural production.

Does Eating Local Reduce Emissions?
with Levi Crews, Ishan Nath, Milan Quentel, and James Sayre

Abstract: This paper examines the conventional wisdom that promoting the consumption of locally produced food reduces greenhouse gas emissions. We first quantify the consequences of an individual consumer's food-sourcing decisions for the carbon footprint of their diet. We combine spatially explicit estimates of production emissions for 46 agricultural products with a model of global transportation and processing

networks, allowing us to calculate the average supply-chain emissions associated with sourcing each product from alternative locations worldwide. We show that the spatial variation in emissions from producing food products is large relative to the emissions from shipping them, and in many cases, consumers minimize emissions by purchasing food from a non-local source. We then use a quantitative spatial model to investigate the aggregate emissions implications of global trade in agricultural goods.

Growth Beneath the Surface: Trade, Groundwater, and Agricultural Development

Publications

Do California's Cattle Emit Less Carbon Than We Think? Published in *ARE Update* in 2025 with James Sayre.

Research Experience

Graduate Student Researcher, UC Davis

with James Sayre 2024–2025
with Michael Carter 2023–2024

Teaching Experience

Teaching Assistant, UC Davis

Econometrics (M.S. level) with Timothy Beatty — Fall 2026
Operations Science with Bulat Gafarov — Spring 2026
Agricultural Markets with Timothy Beatty — Winter 2026
Economics of Business Regulation with Katrina Jessoe — Fall 2025
Development Economics with Warda Riaz — Summer 2023
Intermediate Microeconomics with Kevin Novan — Fall 2022

Guest Lecturer

Econometrics (M.S. level), UC Davis — Nov 2026
Issues and Opportunities in Economics (first-year seminar), University of Maine — Oct 2026

Fellowships and Awards

2024, 2026 Agricultural and Resource Economics Summer Research Fellowship, UC Davis
2024–2026 Jastro-Shields Graduate Research Award, UC Davis
2024 Giannini Foundation Mini-Grant (with James Sayre)
2023 Non-Resident Tuition Fellowship, UC Davis
2019 Lavery Scholarship for Excellence in Economics, University of Maine

Professional Activities

Conference and Seminar Presentations

2026 UC Davis Environmental and Energy Economics Seminar (scheduled, Dec 2026)
2021 Northeast Agricultural and Resource Economics Association Annual Meeting

Service

2024–2025 UC Davis Agricultural Economics Workshop Student Coordinator

2023–2024 Graduate Student Association Representative

Media and Outreach

Extreme Reactions to Extreme Events: The Case of Rice Export Restrictions. Ag Data News, with Greg Boudreaux (2024).

Why do we toss 33% of food we produce when people are hungry? The Maine Question Podcast (2020).

Technical Reports

Bid Inflation for Highway and Bridge Projects: Challenges & Solutions. Prepared for the Maine Department of Transportation (2022).

Other Experience

Short-Term Research Consultant, Margaret Chase Smith Policy Center 2022

(Updated November 4, 2026)